

WEEK 4 OVERVIEW

An aerial photograph of a residential neighborhood that has been completely inundated with floodwater. The water is a murky, brownish-grey color, covering the streets, yards, and surrounding areas. Numerous houses with various roof colors (brown, grey, red) are visible, partially submerged. Green trees are scattered throughout the landscape, some with their lower branches underwater. A central street runs vertically through the middle of the image, with a few vehicles visible in the water. In the upper right corner, a green golf course is partially visible, also surrounded by floodwater. The overall scene depicts a severe flooding event.

James Doss-Gollin

CEVE 543

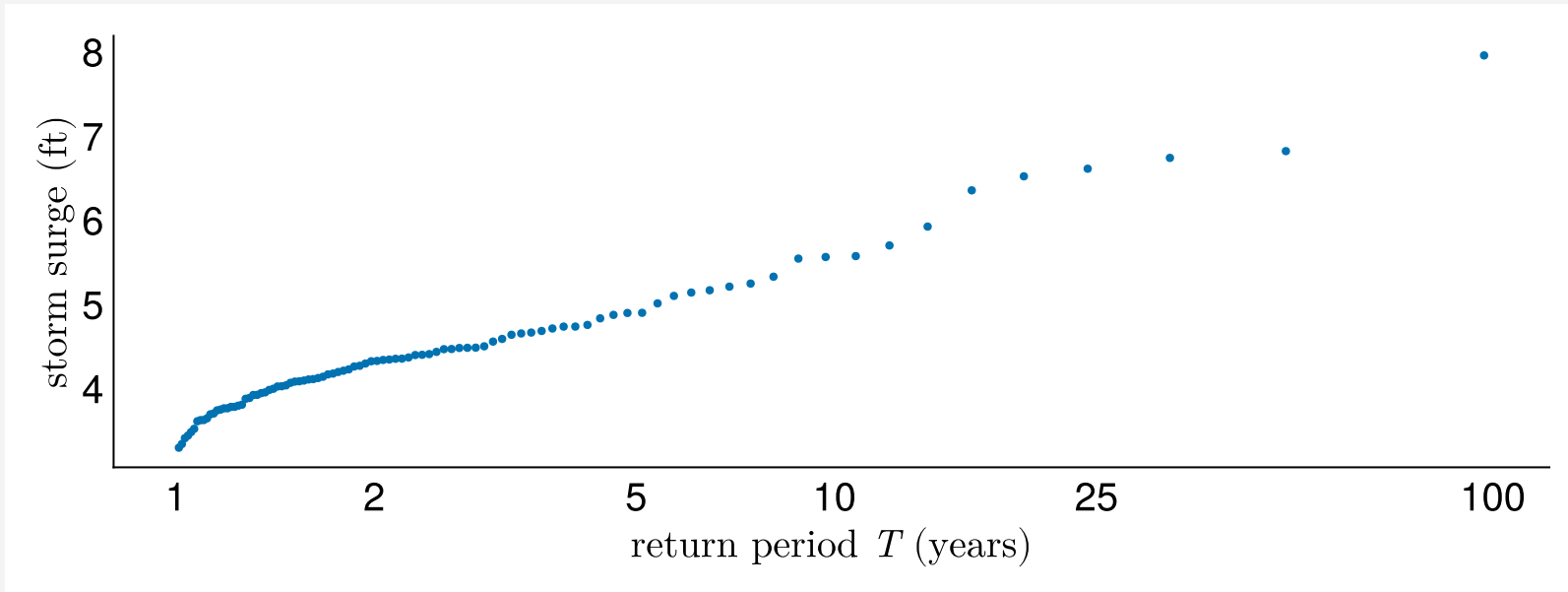
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Module 2: univariate extremes

HOW DO WE CHARACTERIZE A SINGLE HAZARD AT A SINGLE LOCATION?

How do we characterize the probability distribution of a single hydroclimate hazard at a single location, and what are the associated uncertainties?

WE HAVE ALREADY WORKED WITH ANNUAL MAXIMA



Sewells Point annual maximum storm surge, 1928-2023, at Weibull plotting positions. The largest observation gets a return period of about 97 years. What is the 200-year level? The 500-year level?

EXTREME VALUE THEORY GIVES US A PRINCIPLED ANSWER

- Extrapolating beyond the data requires a model
- We would like a justification beyond “it fits”

Extreme value theory to the rescue: a limit theorem says that block maxima converge to a specific family of distributions (the GEV).

WHAT THIS WEEK COVERS

- **Monday:** the GEV distribution and how to estimate its parameters
- **Wednesday:** fitting a GEV in Julia, diagnostics, and Bayesian inference
- **Friday:** lab 4, writing your own model in Turing and testing its sensitivity